

Ethical and Social Challenges in Virtual Reality: A Bibliometric Analysis

Jorge Buele
Carrera de Ingeniería en
Tecnologías de la Información
Universidad Tecnológica
Indoamérica
Ambato, Ecuador
jorgebuele@uti.edu.ec

Homero J. Velasteguí
Carrera de Ingeniería en
Tecnologías de la Información
Pontificia Universidad Católica
del Ecuador
Quito, Ecuador
homero.j.velastegui.i@pucese.edu.ec

Fátima Avilés-Castillo
Carrera de Ingeniería en
Tecnologías de la Información
Universidad Tecnológica
Indoamérica
Ambato, Ecuador
faviles@indoamerica.edu.ec

Hugo Paz-León
Carrera de Ingeniería en
Tecnologías de la Información
Universidad Nacional de
Chimborazo
Riobamba, Ecuador
hpaz@unach.edu.ec

Guillermo Palacios-Navarro
Department of Electronic
Engineering and
Communications
University of Zaragoza
Teruel, Spain
guillermo.palacios@unizar.es

Abstract— Virtual reality has revolutionized human interaction with digital environments, creating immersive simulations that have significantly impacted sectors such as education, urban planning, and mental health. Along with its transformative potential, VR presents ethical and societal challenges. This study analyzes trends in research addressing these challenges, focusing on articles published through 2023 in the Web of Science database. Visual and statistical analyses reveal a sustained increase in publications since 2013. Patterns in authorship and collaboration networks emphasize the global and interdisciplinary nature of VR research. Dominant themes include the metaverse and therapeutic applications, as well as concerns about data privacy, virtual identity representation, and psychological effects such as cybersickness. Research highlights temporary symptoms such as nausea and eye strain associated with immersive systems, demonstrating the need for ergonomic design and age-specific usage guidelines. Furthermore, active video games using VR have shown promise in promoting physical activity, offering innovative solutions to combat sedentary lifestyles. Looking ahead, efforts must prioritize equitable access, user safety, and robust ethical frameworks to ensure the responsible and sustainable integration of VR technologies.

Keywords— *Virtual reality; ethical challenges; social challenges; bibliometric analysis.*

I. INTRODUCTION

Virtual Reality (VR) revolutionized interaction with digital environments, creating immersive three-dimensional simulations that profoundly transformed the connection with the digital world.[1] This technology demonstrated its relevance in education, entertainment, and health, offering previously unreachable experiences [2]. VR allows users to experience environments in a vivid and realistic manner, with significant applications in professional training, psychological therapy, and interactive learning [3], [4]. This advancement requires an ethical framework that considers its social and psychological impacts [5].

The use of VR introduced ethical and social challenges that require constant evaluation. Data privacy is a growing concern, especially in the collection of sensitive information [6]. Virtual identity representation raises issues of authenticity and manipulation, while psychological effects like cyberaddiction and depersonalization demand thorough examination. Prolonged use can distort reality perception, impacting mental health and social interactions, highlighting the need for regulation. VR can foster dependence by isolating users from the real world. This balance between connectivity benefits and risks of digital isolation is vital. Additionally, unequal access to these technologies can widen social and economic gaps, necessitating inclusive policies [7].

As VR becomes more prevalent in daily life, understanding its effects on social behavior is crucial. Studies have examined VR's social applications, emphasizing its advantages and risks, but further research is essential. Long-term impact evaluations will aid in creating policies and regulations to protect users and encourage responsible development. This bibliometric study analyzes current trends in VR's ethical and social challenges using data from the Web of Science (WoS) database to highlight key metrics, emerging trends, and research gaps.

II. METHODOLOGY

A. Data Sources and Search Strategy

This study utilized data from the Web of Science (WoS) database. The search terms included "ethical," "social," "challenges," "virtual," and "reality," combined with Boolean operators. Inclusion criteria were: (i) articles in English, (ii) published up to December 31, 2023, and (iii) all document types (e.g., journal articles, conference papers, reviews, notes, and letters). After applying these criteria, 74 articles were selected based on their titles, abstracts, and keywords, focusing on VR's ethical and social challenges. Irrelevant articles were excluded. The selected articles, along with their full records and cited references, were exported in CSV and plain text formats for analysis.

B. Data Analysis

VOSviewer software was employed for visual bibliometric analysis, enabling the examination of author and journal distributions as well as co-cited references. It also generated visualizations illustrating relationships between keywords. Co-occurrence maps, co-authorship networks, and citation diagrams identified key trends. Descriptive charts highlighting emerging research areas in VR's ethical and social challenges were created using Bibliometrix and Python, which facilitated graphical representations.

C. Research Questions

The research questions guiding this study are as follows:

RQ1: How has research on ethical and social challenges in virtual reality evolved in terms of publication and citation trends?

RQ2: What are the most influential articles in the research on this theme, and what topics do they address?

RQ3: What are the most recurrent keywords in the literature on ethical and social challenges of virtual reality?

RQ4: Who are the most prolific authors and what collaboration networks exist?

RQ5: What are the main journals and conferences publishing research on ethical and social challenges of virtual reality, and what is their impact in terms of citations?

RQ6: How do these research align with the Sustainable Development Goals and what themes are most represented?

III. RESULTS

A. Research Evolution

The annual publication rate indicates the evolution in research on ethical and social challenges of virtual reality. Fig. 1 shows the annual distribution of publications over the last two decades. Despite fluctuations, an increase in annual publications is observed. Since 2013, there has been a notable rise in interest, coinciding with the boom in extended reality environments. Since 2019, it appears to have stabilized at around 70 publications annually, and this trend is expected to continue. A significant increase in citations also indicates the growing relevance of the field, reaching its peak in 2022.

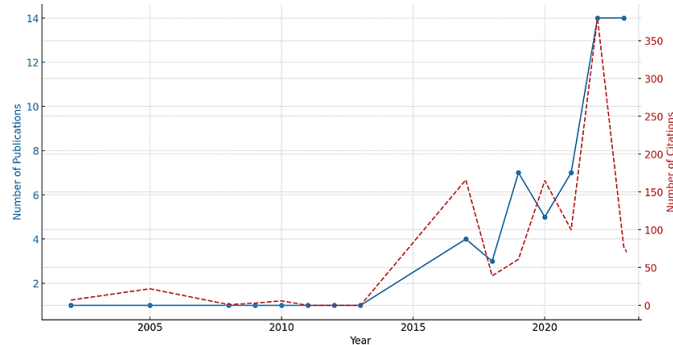


Fig. 1. Scientific production and citations by year.

B. Most Influential Publications

The most influential articles demonstrate significant academic impact (Table 1). Notably, two 2022 studies, "The Metaverse as a Virtual Form of Smart Cities" and "The Social Shaping of the Metaverse as an Alternative to the Imaginaries of Data-Driven Smart Cities," have garnered 135 and 235 citations, with annual rates of 67.5 and 117.5, respectively. This highlights the growing importance of the metaverse and smart cities in discussions on sustainability and urban technology, emphasizing ethical and sustainable development concerns.

Another key study, "Could Virtual Reality Applications Pose Real Risks to Children and Adolescents?", shows an annual citation rate of 82.5, reflecting concerns about VR's impact on youth. Similarly, "Deep and Meaningful E-Learning with Social Virtual Reality Environments in Higher Education" illustrates VR's potential in education, with 38 annual citations. These studies reveal diverse VR applications, from education to mental health, underscoring the need for ongoing research into ethical and social implications.

C. Trend Analysis

Fig. 2 illustrates the keyword network, with "virtual reality" branching into themes such as "metaverse" and "augmented reality" in urban and sustainable contexts. Terms like "therapy" and "anxiety" highlight therapeutic applications, while "children" and "education" reflect educational impacts. Keywords like "privacy" and "ethics" emphasize privacy concerns, linking VR with "artificial intelligence" and "smart cities".

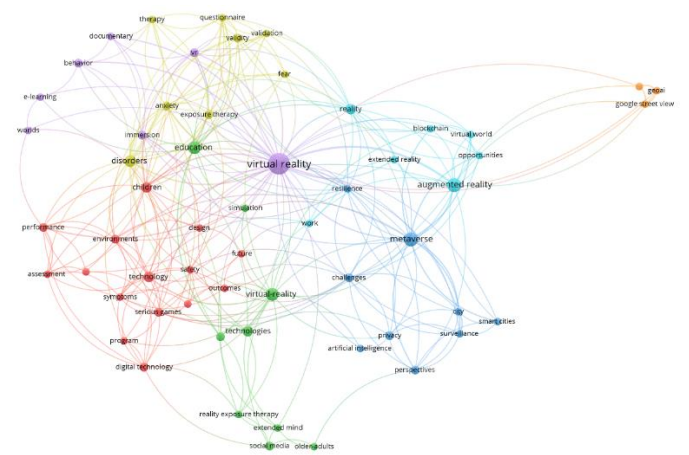


Fig. 2. Scientific production and citations by year.

Fig. 3 shows a thematic network that categorizes themes into four quadrants based on their development density and centrality. "Virtual Reality", "Social", and "Education" are in the quadrant of core themes, indicating their central relevance and ongoing development in research. Themes in the driver quadrant, such as "Challenges" and "Technologies", are highly developed and central, driving significant advances in the field. In contrast, themes like "Digital" and "Health" in the quadrant of emerging themes suggest areas of nascent or declining research. This visualization provides a clear view of the current priorities and directions in research on virtual reality and its ethical and social applications.

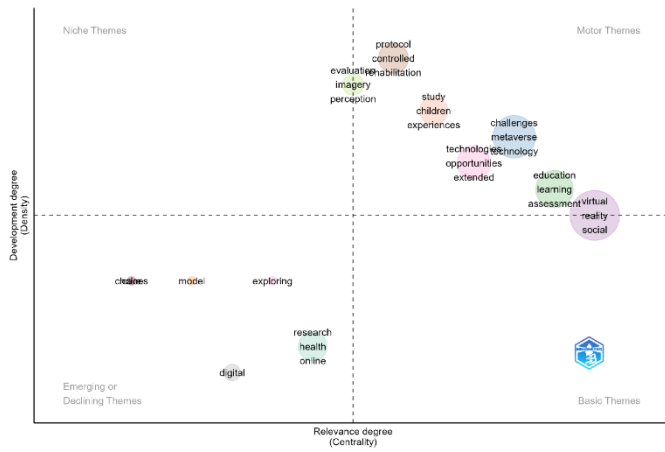


Fig. 3. Thematic map of research on virtual reality and its ethical and social challenges.

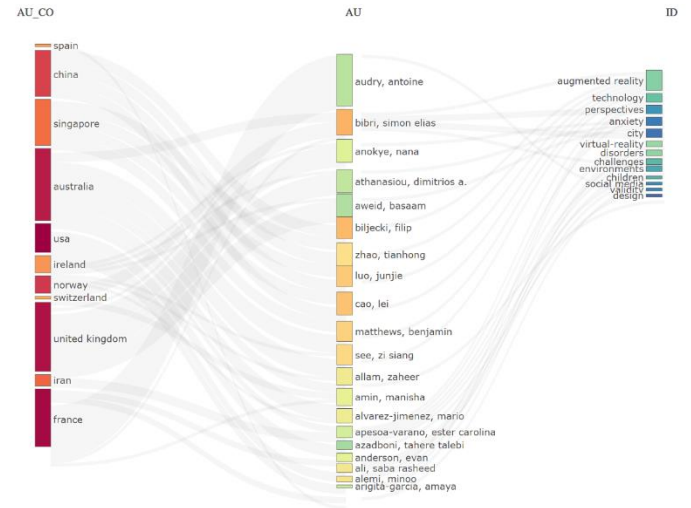


Fig. 4. Connection between countries, authors, and keywords.

D. International Collaboration and Publications

Author Collaboration Networks. Figure 4 illustrates the connections between countries, authors, and keywords in this research field. The United States, United Kingdom, China, and Australia emerge as leading contributors. Authors like Antoine Audry, Simon Elias Bibri, and Nana Anokye stand out for their prominence and central roles, indicating substantial influence. The links between these authors and specific keywords suggest their leadership in studies on virtual and augmented reality within social and ethical contexts. This visualization underscores both international and individual collaborations.

Most Influential Journals. The most influential journals in virtual reality's ethical and social challenges include "Ethical Challenges in Digital Psychology and Cyberpsychology" (Cambridge University Press), with an average of 884 citations per publication, and "Smart Cities" (MDPI), with 205 citations per publication. These journals highlight the focus on sustainable development and urban technologies, emphasizing the value of high-impact outlets for research visibility. Fig. 5

TABLE I. ARTICLES WITH THE MOST CITATIONS

N°	Reference	Article	Journal	Year	Cites	TCPY
1	Allam et al. [8]	The Metaverse as a Virtual Form of Smart Cities: Opportunities and challenges for environmental, economic, and social sustainability in urban futures	Smart Cities	2022	135	67.50 (#3)
2	Haghani and Sarvi [9]	Stated and revealed exit choices of pedestrian crowd evacuees	Transportation Research Part B-Methodological	2017	50	7.14 (#8)
3	Bell et al. [10]	Virtual reality as a clinical tool in mental health research and practice	Dialogues In Clinical Neuroscience	2020	84	21.00 (#6)
4	Han et al. [11]	Virtual reality consumer experience escapes: preparing for the metaverse	Virtual Reality	2022	88	44.00 (#4)
5	Lindeman et al. [12]	Technology and Caregiving: Emerging Interventions and Directions for Research	Gerontologist	2020	59	14.75 (#7)
6	Mystakidis et al. [13]	Deep and Meaningful E-Learning with Social Virtual Reality Environments in Higher Education: A Systematic Literature Review	Applied Sciences-Basel	2021	114	38.00 (#5)
7	Kaimara et al. [14]	Could virtual reality applications pose real risks to children and adolescents? A systematic review of ethical issues and concerns	Virtual Reality	2022	165	82.50 (#2)
8	Bibri [15]	The Social Shaping of the Metaverse as an Alternative to the Imaginaries of Data-Driven Smart Cities: A Study in Science, Technology, and Society	Smart Cities	2022	235	117.50 (#1)
9	McArthur and Twist [16]	The rise of digisexuality: therapeutic challenges and possibilities	Sexual and Relationship Therapy	2017	30	4.29 (#9)
10	Kenwright [17]	Virtual Reality: Ethical Challenges and Dangers Physiological and Social Impacts	IEEE Technology and Society Magazine	2018	25	4.17 (#10)

illustrates the connections between authors' countries, journals, and keywords.

TABLE II. MAIN JOURNALS FOR PUBLISHING RESEARCH ARTICLES.

Editorial	Journal	Publication	Citations	TCPA
MDPI	Smart Cities	3	615	205 (#2)
Cambridge Univ Press	Ethical Challenges in Digital Psychology and Cyberpsychology	2	1768	884 (#1)
Springer London Ltd	Virtual Reality	2	253	126,5 (#6)
JMIR Publications, Inc	Journal Of Medical Internet Research	2	172	86 (#8)
JMIR Publications, Inc	Jmir Research Protocols	2	142	71 (#9)
Frontiers Media Sa	Frontiers In Psychology	2	128	64 (#10)
Sage Publications Inc	Clinical Psychological Science	1	181	181 (#3)
Elsevier	Landscape And Urban Planning	1	163	163 (#4)
MDPI	Civileng	1	137	137 (#5)
MDPI	Journal Of Clinical Medicine	1	126	126 (#7)

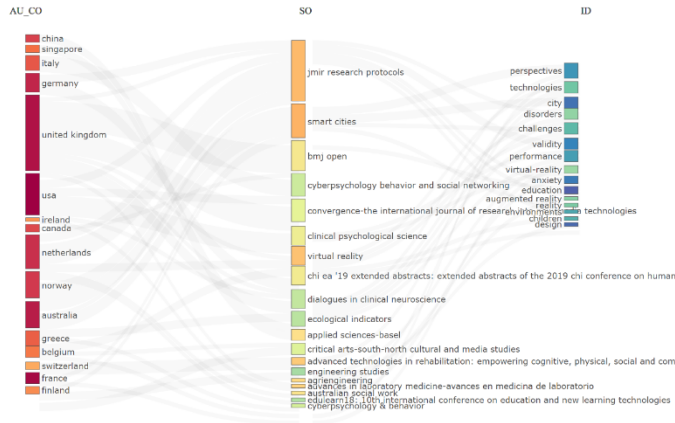


Fig. 5. Connection between countries of authors, journals, and keywords.

E. Themes in the Context of SDGs

In Fig. 6 displays the distribution of articles and their relevance to the Sustainable Development Goals (SDGs) in the study of ethical and social aspects of virtual reality. The most represented goal is "Quality Education", encompassing more than 40% of the records, highlighting the priority of education in research related to VR. "Good Health and Well Being" is also prominent, comprising about 20% of the records, reflecting the importance of this technology in promoting health. Other highlighted themes include "Sustainable Cities and Communities" and "Gender Equality", indicating a strong alignment of research on ethical and social challenges of VR with the SDGs.

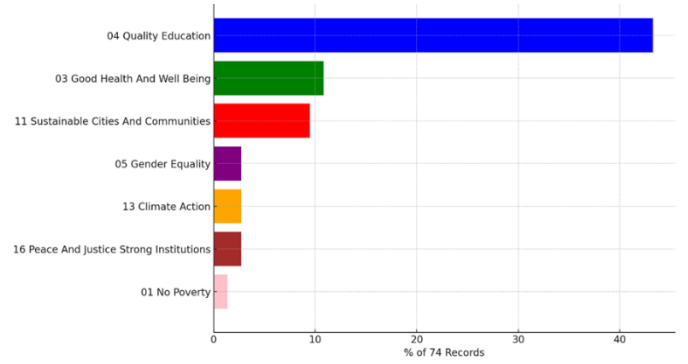


Fig. 6. Distribution of SDG themes in the dataset.

IV. DISCUSSION

International collaboration networks significantly impact VR research, with countries like the United States, United Kingdom, China, and Australia leading in scientific output and partnerships [18]. These collaborations enhance the quality and quantity of VR publications while fostering interdisciplinary approaches to address VR's social and ethical challenges. A collaborative framework allows researchers to analyze VR's implications from multiple perspectives, deepening understanding of its benefits and risks.

VR has transformative potential in urban planning and mental health. Digital twins and metaverse simulations in urban design enable policies to be visualized and tested in realistic environments before implementation, supporting sustainable development [19]. Active video games integrated with VR encourage physical activity and provide innovative methods to combat childhood obesity and promote healthier habits [20]. In mental health, VR creates immersive settings for treatments like exposure therapy and anxiety management, offering new therapeutic options. However, challenges such as cybersickness, marked by symptoms like nausea and dizziness are more common with immersive systems than traditional displays [21]. Advances in hardware and software have improved usability, even for vulnerable groups like the elderly [22].

Ethical considerations, particularly regarding data privacy and virtual identity representation, are significant as VR technology collects extensive personal and behavioral data, raising privacy concerns in fields like healthcare and education. Robust regulatory frameworks are needed to protect users while supporting innovation. For example, studies on immersive VR's effects on children's visual development indicate that while long-term impacts are minimal, temporary issues like eyestrain and motion sickness require age-appropriate guidelines and hardware improvements to ensure safety and usability [23].

In education, VR enhances learning by creating immersive experiences that improve retention and understanding, particularly in technical training where simulations provide safe, practical environments [24]. To increase accessibility, inclusive policies and ethical guidelines are essential for VR's broader application. Future research must employ rigorous methods to ensure VR's benefits extend to all socioeconomic

groups and address long-term social implications, encouraging ethical and equitable use across sectors [25].

V. CONCLUSIONS

This study highlights the growing focus on virtual reality (VR) as a transformative technology in areas such as the metaverse, augmented reality, and mental health. International collaborations, particularly involving researchers from the United States, United Kingdom, China, and Australia, have contributed to advancements in fields like urban planning, education, and therapeutic applications. Future research should prioritize addressing global challenges such as accessibility, equity, and safety in VR applications. Emphasis on ethical and inclusive methodologies is essential to ensure VR technologies enhance societal well-being while minimizing risks, particularly in data privacy and virtual identity representation. Developing robust regulatory frameworks and interdisciplinary approaches will support the responsible integration of VR across diverse sectors, fostering sustainable and equitable technological progress.

ACKNOWLEDGMENT

Special thanks to Universidad Tecnológica Indoamérica for its support through the project IIDI-010-24, titled “Exploración de los parámetros inherentes a los entornos virtuales para la estimulación cognitiva” and to the EDUTEM research network for its valuable contributions..

REFERENCES

- [1] Z. Lv, «Virtual reality in the context of Internet of Things», *Neural Comput & Applic*, vol. 32, n.º 13, pp. 9593-9602, jul. 2020, doi: 10.1007/s00521-019-04472-7.
- [2] Z. Andrea Sánchez *et al.*, «Virtual Rehabilitation System Using Electromyographic Sensors for Strengthening Upper Extremities», en *Developments and Advances in Defense and Security*, Á. Rocha y R. P. Pereira, Eds., Singapore: Springer, 2020, pp. 231-241. doi: 10.1007/978-981-13-9155-2_19.
- [3] J. Buele, J. L. Varela-Aldás, y G. Palacios-Navarro, «Virtual reality applications based on instrumental activities of daily living (iADLs) for cognitive intervention in older adults: a systematic review», *J NeuroEngineering Rehabil*, vol. 20, n.º 1, p. 168, dic. 2023, doi: 10.1186/s12984-023-01292-8.
- [4] M. Campaña-Cordova, F. Castillo-Ledesma, J. Oleas-Orozco, H. Moncayo, y F. Avilés-Castillo, «Enhancing the Teaching of Natural Sciences in Rural Environments through Educational Videos», en *2023 IEEE Seventh Ecuador Technical Chapters Meeting (ECTM)*, oct. 2023, pp. 1-5. doi: 10.1109/ETCM58927.2023.10309022.
- [5] J. Buele, F. Avilés-Castillo, C. Del-Valle-Soto, J. Varela-Aldás, y G. Palacios-Navarro, «Effects of a dual intervention (motor and virtual reality-based cognitive) on cognition in patients with mild cognitive impairment: a single-blind, randomized controlled trial», *J NeuroEngineering Rehabil*, vol. 21, n.º 1, p. 130, ago. 2024, doi: 10.1186/s12984-024-01422-w.
- [6] Y. Wang *et al.*, «A Survey on Metaverse: Fundamentals, Security, and Privacy», *IEEE Communications Surveys & Tutorials*, vol. 25, n.º 1, pp. 319-352, 2023, doi: 10.1109/COMST.2022.3202047.
- [7] M. Pilatásig *et al.*, «Interactive System for Hands and Wrist Rehabilitation», en *Proceedings of the International Conference on Information Technology & Systems (ICITS 2018)*, Á. Rocha y T. Guarda, Eds., Cham: Springer International Publishing, 2018, pp. 593-601. doi: 10.1007/978-3-319-73450-7_56.
- [8] Z. Allam, A. Sharifi, S. E. Bibri, D. S. Jones, y J. Krogstie, «The Metaverse as a Virtual Form of Smart Cities: Opportunities and Challenges for Environmental, Economic, and Social Sustainability in Urban Futures», *Smart Cities*, vol. 5, n.º 3, Art. n.º 3, sep. 2022, doi: 10.3390/smartcities5030040.
- [9] M. Haghighi y M. Sarvi, «Stated and revealed exit choices of pedestrian crowd evacuees», *Transportation Research Part B: Methodological*, vol. 95, pp. 238-259, ene. 2017, doi: 10.1016/j.trb.2016.10.019.
- [10] I. H. Bell, J. Nicholas, M. Alvarez-Jimenez, A. Thompson, y L. Valmaggia, «Virtual reality as a clinical tool in mental health research and practice», *Dialogues in Clinical Neuroscience*, vol. 22, n.º 2, pp. 169-177, jun. 2020, doi: 10.31887/DCNS.2020.22.2/lvalmaggia.
- [11] D.-I. D. Han, Y. Bergs, y N. Moorhouse, «Virtual reality consumer experience escapes: preparing for the metaverse», *Virtual Reality*, vol. 26, n.º 4, pp. 1443-1458, dic. 2022, doi: 10.1007/s10055-022-00641-7.
- [12] D. A. Lindeman, K. K. Kim, C. Gladstone, y E. C. Apesoa-Varano, «Technology and Caregiving: Emerging Interventions and Directions for Research», *The Gerontologist*, vol. 60, n.º Supplement_1, pp. S41-S49, feb. 2020, doi: 10.1093/geront/gnz178.
- [13] S. Mystakidis, E. Berki, y J.-P. Valtanen, «Deep and Meaningful E-Learning with Social Virtual Reality Environments in Higher Education: A Systematic Literature Review», *Applied Sciences*, vol. 11, n.º 5, Art. n.º 5, ene. 2021, doi: 10.3390/app11052412.
- [14] P. Kaimara, A. Oikonomou, y I. Deliyannis, «Could virtual reality applications pose real risks to children and adolescents? A systematic review of ethical issues and concerns», *Virtual Reality*, vol. 26, n.º 2, pp. 697-735, jun. 2022, doi: 10.1007/s10055-021-00563-w.
- [15] S. E. Bibri, «The Social Shaping of the Metaverse as an Alternative to the Imaginaries of Data-Driven Smart Cities: A Study in Science, Technology, and Society», *Smart Cities*, vol. 5, n.º 3, Art. n.º 3, sep. 2022, doi: 10.3390/smartcities5030043.
- [16] N. McArthur y M. L. C. Twist, «The rise of digisexuality: therapeutic challenges and possibilities», *Sexual and Relationship Therapy*, vol. 32, n.º 3-4, pp. 334-344, oct. 2017, doi: 10.1080/14681994.2017.1397950.
- [17] B. Kenwright, «Virtual Reality: Ethical Challenges and Dangers», *IEEE Technology and Society*, ene. 2019, doi: 10.1109/MTS.2018.2876104.
- [18] B. T. Familoni y N. C. Onyebuchi, «Augmented and Virtual Reality in U.S. Education: A Review: Analyzing The Impact, Effectiveness, and Future Prospects of Ar/Vr Tools in Enhancing Learning Experiences», *International Journal of Applied Research in Social Sciences*, vol. 6, n.º 4, Art. n.º 4, abr. 2024, doi: 10.51594/ijarss.v6i4.1043.
- [19] J. Lin y M. E. Latoschik, «Digital body, identity and privacy in social virtual reality: A systematic review», *Front. Virtual Real.*, vol. 3, nov. 2022, doi: 10.3389/frvir.2022.974652.
- [20] I. A. Ramírez-Granizo, J. L. Ubago-Jiménez, G. González-Valero, P. Puertas-Molero, y S. San Román-Mata, «The Effect of Physical Activity and the Use of Active Video Games: Exergames in Children and Adolescents: A Systematic Review», *International Journal of Environmental Research and Public Health*, vol. 17, n.º 12, Art. n.º 12, ene. 2020, doi: 10.3390/ijerph17124243.
- [21] L. Rebenitsch y C. Owen, «Review on cybersickness in applications and visual displays», *Virtual Reality*, vol. 20, n.º 2, pp. 101-125, jun. 2016, doi: 10.1007/s10055-016-0285-9.
- [22] B. F. Drazich *et al.*, «In too deep? A systematic literature review of fully-immersive virtual reality and cybersickness among older adults», *Journal of the American Geriatrics Society*, vol. 71, n.º 12, pp. 3906-3915, 2023, doi: 10.1111/jgs.18553.
- [23] L. Tychsen y P. Foeller, «Effects of Immersive Virtual Reality Headset Viewing on Young Children: Visuomotor Function, Postural Stability, and Motion Sickness», *American Journal of Ophthalmology*, vol. 209, pp. 151-159, ene. 2020, doi: 10.1016/j.ajo.2019.07.020.
- [24] S. Olbina y S. Glick, «Using Integrated Hands-on and Virtual Reality (VR) or Augmented Reality (AR) Approaches in Construction Management Education», *International Journal of Construction Education and Research*, vol. 19, n.º 3, pp. 341-360, jul. 2023, doi: 10.1080/15578771.2022.2115173.
- [25] E. Dhar *et al.*, «A scoping review to assess the effects of virtual reality in medical education and clinical care», *DIGITAL HEALTH*, vol. 9, p. 20552076231158022, ene. 2023, doi: 10.1177/20552076231158022.